

Exercises: Built-in Functions

You can check your solutions in the [Judge system](#).

Part I – Queries for SoftUni Database

1. Find Names of All Employees by First Name

Create a SQL query that finds all employees whose **first name starts with "Sa"**. As a result, display **"FirstName"** and **"LastName"**.

Example

FirstName	LastName
Sariya	Harnpadoungsataya
Sandra	Reategui Alayo
...	...

2. Find Names of All Employees by Last Name

Create a SQL query that finds all employees whose **last name contains "ei"**. As a result, display **"FirstName"** and **"LastName"**.

Example

FirstName	LastName
Kendall	Keil
Christian	Kleinerman
...	...

3. Find First Names of All Employees

Create a SQL query that finds the **first names** of all employees whose **department ID** is **3** or **10**, and the **hire year** is **between 1995 and 2005 inclusive**.

Example

FirstName
Deborah
Wendy
Candy
...

4. Find All Employees Except Engineers

Create a SQL query that finds the **first and last names** of every employee, whose **job title does not contain "engineer"**.

Example

FirstName	LastName
Guy	Gilbert
Kevin	Brown
Rob	Walters

...	...
-----	-----

5. Find Towns with Name Length

Create a SQL query that finds all town names, which are **5 or 6 symbols long**. Order the result **alphabetically by town name**.

Example

Name
Berlin
Duluth
Duvall
...

6. Find Towns Starting With

Create a SQL query that finds all towns with names **starting with 'M', 'K', 'B' or 'E'**. Order the result **alphabetically by town name**.

Example

TownID	Name
5	Bellevue
31	Berlin
30	Bordeaux
...	...

7. Find Towns Not Starting With

Create a SQL query that finds all towns, which **do not start with 'R', 'B' or 'D'**. Order the result **alphabetically by name**.

Example

TownID	Name
2	Calgary
23	Cambridge
15	Carnation
...	...

8. Create View Employees Hired After 2000 Year

Create a SQL query that creates view "**V_EmployeesHiredAfter2000**" with **the first and the last name** for all employees, **hired after the year 2000**.

Example

FirstName	LastName
Steven	Selikoff
Peter	Krebs
Stuart	Munson
...	...

9. Length of Last Name

Create a SQL query that finds **all employees**, whose **last name is exactly 5 characters long**.

Example

FirstName	LastName
Kevin	Brown
Terri	Duffy
Jo	Brown
Diane	Glimp
...	...

10. Rank Employees by Salary

Write a query that **ranks** all employees using **DENSE_RANK**. In the **DENSE_RANK** function, employees need to be **partitioned** by **Salary** and **ordered** by **EmployeeID**. You need to find **only** the employees, whose **Salary** is between **10000** and **50000** and **order** them by **Salary** in **descending order**.

Example

EmployeeID	FirstName	LastName	Salary	Rank
268	Stephen	Jiang	48100.00	1
284	Amy	Alberts	48100.00	2
288	Syed	Abbas	48100.00	3
...	...	...	...	...

11. Find All Employees with Rank 2



Upgrade the query from the previous problem, so that it finds only the employees with a **Rank 2**. **Order** the result by **Salary (descending)**.

Example

EmployeeID	FirstName	LastName	Salary	Rank
284	Amy	Alberts	48100.00	2
292	Martin	Kulov	48000.00	2
71	Wendy	Kahn	43300.00	2
...	...	...	...	...

Part II – Queries for Geography Database

12. Countries Holding 'A' 3 or More Times

Find all countries which hold the letter 'A' at least 3 times in their name (case-insensitively). **Sort** the **result by ISO code** and display the "Country Name" and "ISO Code".

Example

Country Name	ISO Code
Afghanistan	AFG
Albania	ALB
...	...

13. Mix of Peak and River Names



Combine all peak names with all river names, so that the **last letter** of each **peak name** is the **same as** the **first letter** of its corresponding **river name**. Display the **peak names**, **river names** and the **obtained mix** (mix should be in lowercase). **Sort** the results **by** the **obtained mix**.

Example

PeakName	RiverName	Mix
Aconcagua	Amazon	aconcaguamazon
Aconcagua	Amur	aconcaguamur
Banski Suhodol	Lena	banski suhodolena
...	...	...

Part III – Queries for Diablo Database

14. Games from 2011 and 2012 Year

Find and display the **top 50 games** which start date is from **2011** and **2012** year. **Order** them by start date, **then by name of the game**. The start date should be in the following format: "yyyy-MM-dd".

Example

Name	Start
Rose Royalty	2011-01-05
London	2011-01-13
Broadway	2011-01-16
...	...

15. User Email Providers

Find all **users** with information about their **email providers**. Display the **username** and **email provider**. **Sort** the results by **email provider alphabetically**, **then by username**.

Example

Username	Email Provider
Pesho	abv.bg
monoxidecos	astonrasuna.com
bashsassafras	balibless
...	...

16. Get Users with IP Address Like Pattern



Find all **users** with their **IP addresses**, sorted by **username alphabetically**. Display only the rows which IP address matches the pattern: `"***.1^.^.*"`.

Legend:

* - one symbol

^ - one or more symbols

Example

Username	IP Address
bindbawdy	192.157.20.222
evolvingimportant	223.175.227.173
inguinalself	255.111.250.207
...	...

17. Show All Games with Duration and Part of the Day

Find all **games** with **part of the day** and **duration**. Sort them **by game name** alphabetically, **then by duration** (alphabetically, not by the timespan) and **part of the day** (all ascending). **Part of the Day** should be **Morning** (time is ≥ 0 and < 12), **Afternoon** (time is ≥ 12 and < 18), **Evening** (time is ≥ 18 and < 24). **Duration** should be **Extra Short** (smaller or equal to 3), **Short** (between 4 and 6 including), **Long** (greater than 6) and **Extra Long** (without duration).

Example

Game	Part of the Day	Duration
Ablajeck	Morning	Long
Ablajeck	Afternoon	Short
Abregado Rae	Afternoon	Long
Abrion	Morning	Extra Short
Acaeria	Evening	Long
...	...	...

Part IV – Date Functions Queries

18. Orders Table



You are given a table **Orders(Id, ProductName, OrderDate)** filled with data. Consider that the **payment** for that order must be accomplished **within 3 days after the order date**. Also the **delivery date is up to 1 month**. Write a query to show each product's **name, order date, pay and deliver due dates**.

Original Table

Id	ProductName	OrderDate
1	Butter	2016-09-19 00:00:00.000
2	Milk	2016-09-30 00:00:00.000
3	Cheese	2016-09-04 00:00:00.000
4	Bread	2015-12-20 00:00:00.000
5	Tomatoes	2015-01-01 00:00:00.000
...	...	...

Output

ProductName	OrderDate	Pay Due	Deliver Due
Butter	2016-09-19 00:00:00.000	2016-09-22 00:00:00.000	2016-10-19 00:00:00.000
Milk	2016-09-30 00:00:00.000	2016-10-03 00:00:00.000	2016-10-30 00:00:00.000
Cheese	2016-09-04 00:00:00.000	2016-09-07 00:00:00.000	2016-10-04 00:00:00.000
Bread	2015-12-20 00:00:00.000	2015-12-23 00:00:00.000	2016-01-20 00:00:00.000
Tomatoes	2015-01-01 00:00:00.000	2016-01-02 00:00:00.000	2016-01-30 00:00:00.000
...	...	...	...

19. People Table



Create a table **People(Id, Name, Birthdate)**. Write a query to **find age in years, months, days and minutes** for each person for the **current time** of executing the query.

Original Table

Id	Name	Birthdate
1	Victor	2000-12-07 00:00:00.000
2	Steven	1992-09-10 00:00:00.000
3	Stephen	1910-09-19 00:00:00.000
4	John	2010-01-06 00:00:00.000
...	...	...

Example Output

Name	Age in Years	Age in Months	Age in Days	Age in Minutes
Victor	16	189	5754	8286787
Steven	24	288	8764	12621187
Stephen	106	1272	38706	55737667
John	6	80	2437	3510307
...	...	...	...	...